PHYS7101 · NUMERICAL METHODS AND DATA ANALYSIS

Exercise 2: Heart rate and breathing — how are they related?

Weight	25% of the module	**Due**	4pm, Friday 20 November 2026, on Moodle
Covers	Lectures 5–8 (and uses interpolation and integration from Lecture 4)	**Submit**	Notebook (already run), ending with your statement (≤ 300 words, up to one figure) + GenAI declaration

1. The data

The *Fantasia* database on PhysioNet holds recordings from 40 healthy adults: 20 young (aged 21–34) and 20 elderly (aged 68–85). Each lay at rest for two hours, watching Disney's *Fantasia* to stay awake, while their ECG (the heart's electrical signal) and their breathing were recorded 250 times a second (333 for one person). Every heartbeat in the ECG has been located automatically; a few may be missed or false. A couple of people have many extra (ectopic) beats: real beats, but not part of the normal rhythm. One person has a short gap in the breathing signal (NaN). Decide what to do about each of these, and say why. (If you want to go back to the original ECG, the last cell of the starter notebook loads the raw recordings.)

You have a 30-minute segment for each person: the time of every heartbeat, and the breathing signal resampled to 4 samples per second. Four samples per second is enough for the rhythms studied here, all slower than about 0.5 Hz: a heart-rate series built from the beats carries information only up to about half the beat rate (Lecture 8). To see the shape of each heartbeat or breath you need many more samples per cycle (about ten), so for that you need the raw recordings, which you can load for an extension. Your job is to describe how the average heart rate and the heart-rate dynamics differ between young and elderly people, to find out whether and how the heart-rate dynamics and the breathing dynamics are related, and to decide how far you trust the evidence.

Our assumption. Over each 30-minute recording we assume that the rule governing the cardiorespiratory system does not change in time: the system is autonomous (Lecture 2). Its signals are then stationary (Lecture 6): their statistics, and the relationships between them, do not change over the recording. This is a simplifying assumption for our analysis, not a claim that the physiology is unchanging. Whether, and how far, it affects the results is an interesting question to explore in your extension.

These are real, de-identified measurements, used under an open licence that requires them to be cited: Iyengar *et al.* (1996), *Am. J. Physiol.* **271**, R1078; and Goldberger *et al.* (2000), *Circulation* **101**, e215. Everyone has the same data.

2. Data and starting notebook

Everything is in the module repository, folder coursework/exercise2:

- **exercise2_heart.ipynb**: loads the data and lists the tasks. Open it in Colab, then **File → Save a copy in Drive**.
- **datasets/fantasia_beats.csv**: one row per heartbeat; columns `subject` and `t_s` (time of the beat, s). **datasets/fantasia_resp.csv**: columns `subject`, `t_s` and `resp` (breathing, arbitrary units). **datasets/fantasia_subjects.csv**: `subject` and `group` (young or elderly).

Terms you will need. The RR interval is the time between successive beats. The instantaneous heart rate (IHR) is 60/RR beats per minute (bpm). The RR intervals plotted against time are the heart-rate variability (HRV); the IHR carries the same information, inverted: a long interval is a low rate (Lecture 8). SDNN is the standard deviation of the normal-to-normal (NN) intervals, the RR intervals after artefacts and ectopic beats have been dealt with; RMSSD is the rms of the differences between successive NN intervals. When methods with linear frequency resolution are used, such as the Fourier spectra in this exercise, heart-rate variability is usually split into a low-frequency band (LF, 0.04–0.15 Hz) and a high-frequency band (HF, 0.15–0.40 Hz), which contains the breathing rhythm.

3. Required tasks

Task 1: One person, in detail (Lectures 5 and 8)

1.1 Intervals. For subject f1o07, compute the RR intervals and plot them against time (the HRV) and as a histogram. A few intervals are artefacts (a missed or an extra beat). Write a rule that flags them, for example an interval more than 20% from the median of the 11 intervals centred on it, or more than 5 MAD from that median (Lecture 8). Justify your rule, and report how many intervals it flags and what you do with them.

1.2 Describe. Give the mean heart rate (say whether you use 60 divided by the mean RR, or the mean of the IHR), SDNN and RMSSD, each with an uncertainty. SDNN and RMSSD have no simple formula for their uncertainty, so use a bootstrap (Lecture 5; Week 4 lab): resample, recompute, and take the spread of the results. Would you quote a mean or a median? Successive intervals are not independent: does a bootstrap that resamples single intervals, or $s/\sqrt{N}$, still give the right uncertainty?

1.3 Breathing. Plot the breathing signal and estimate the breathing rate. At rest it is usually 10–25 breaths per minute: check your estimate against the signal.

Report: the two plots; your artefact rule, the number of intervals it flags and what you did with them; the three quantities with uncertainties, and why you trust those uncertainties; the breathing rate.

Task 2: Young and elderly (Lectures 5 and 8; Week 4 lab)

2.1 All 40 people. Apply your Task 1 functions to every person, and build a table with one row per person: at least mean heart rate, SDNN, RMSSD and breathing rate.

2.2 Compare. Plot mean heart rate, SDNN and RMSSD by group, showing every person. Compare the groups with a two-sample t-test and a rank test (Mann–Whitney), and give each difference with a bootstrap confidence interval obtained by resampling people (the difference can be, for example, a Hodges–Lehmann estimate; Lectures 5 and 8). Which test do you trust for each quantity, and why? You are making several tests at once: what does that change?

Report: the table; the comparison figure; the test results and intervals, with your choice of test for each quantity.

Task 3: Rhythms (Lectures 4, 6 and 8; Week 5 lab)

3.1 Even sampling. Heartbeats are not evenly spaced in time. Interpolate the IHR (or the RR intervals; say which) onto a regular grid of 4 samples per second (for example with a cubic spline; Lectures 4 and 8), and plot a short section with the beats to check it.

3.2 Spectra in time, then averaged. For one young and one elderly person: (a) compute the time–frequency representation of the heart-rate series from 3.1 and of breathing with the short-time Fourier transform (Lecture 6), with 40-s windows every 4 s (at 4 samples per second, nperseg=160 and noverlap=144), and plot it (time against frequency, power as colour). Subtract the mean of each series first, or use signal.spectrogram, which removes it in each window; otherwise the zero-frequency term dominates. How does the breathing rhythm change during the 30 minutes? (b) Average the power over time to get the time-averaged spectrum (Welch's method; signal.welch gives the same in one step), and find the breathing peak. How wide is it, and what makes it wide? Names you will meet: the time–frequency representation from the short-time Fourier transform is usually called a spectrogram (scipy.signal.spectrogram); its average over time is Welch's estimate of the power spectrum, or an averaged periodogram (scipy.signal.welch).

3.3 Band powers. Compute each person's time-averaged spectrum with windows long enough for the LF band, which starts at 0.04 Hz. A 40-s window holds less than two of its cycles, so use several, for example five (about 2 minutes), and say why. signal.welch returns power per hertz, so its integral over a band is the variance in that band. Integrate over the LF and HF bands with your own trapezoid or Simpson function (Lecture 4), and add both powers, with units, to your table.

Report: the time–frequency and time-averaged spectrum figures, and what makes the breathing peak wide; your window choices and why; the band powers for all 40 people.

Task 4: Is there a linear statistical relationship between heart rate and breathing, and at what time lag is it strongest? (Lecture 8; Week 5 lab)

4.1 Cross-correlation. For one person, compute the cross-correlation between breathing and the IHR (with the RR intervals the sign is reversed; Lecture 8). Report its largest value and the lag at which it occurs.

4.2 Is it real? Generate a null distribution by repeatedly shifting the breathing signal in time relative to the heart-rate signal (a surrogate-data test). Compare the results from these time-shifted signals with the result from the original, unshifted data, and estimate how often the surrogates give a result equal to or stronger than the original. Repeat for all 40 people.

Report: the cross-correlation figure; the surrogate test for one person and the summary for all 40.

Task 5: Structure across people (Lecture 7; Week 6 lab)

Take at least five quantities from your table. Standardise them, carry out a principal component analysis, and report how much variance each component explains and which quantities dominate the first two. Plot the people on the first two components, coloured by group. Then cluster them with k-means (k = 2), and compare the clusters with the groups.

Report: the variance explained and the principal directions (the weight of each quantity in a component, called loadings in many packages); the component plot; the comparison of clusters with groups, and which quantities separate young from elderly.

4. Extension (Task 6): what comes next?

Look at what Tasks 1–5 have shown you. Decide what the most important open question is, carry out the analysis that answers it, and check that analysis. In the notebook, say **why** you chose it: the reason should come from your results.

You are assessed on how well your choice follows from your results, and on how well it is done and checked, **not** on how many things you try. One well-chosen, well-tested analysis is worth more than several superficial ones.

Some questions that might help (not a list to work through):

- Does a result change with your choices: the artefact rule, the interpolation, the window length?
- Does the relationship between breathing and heart rate change during the 30 minutes? (Cross-correlation in windows, as in the time–frequency representation of Task 3.2.)
- Is a difference between groups really about age, or about something that differs with age, such as breathing rate?
- Is your quoted uncertainty believable? (Bootstrap or surrogates, Lectures 5 and 8.)
- Is the strength of the relationship (the largest cross-correlation) related to breathing rate, or to age group? (Pearson and Spearman correlations with uncertainties, Lecture 7; say what they can and cannot tell you about cause.)
- Would another method answer your question more directly? For example, coherence (how strongly two signals oscillate together at each frequency), or a spectrum computed directly from the uneven beat times (the Lomb–Scargle periodogram), which needs no interpolation.
- Would the raw recordings show something the 4-sample-per-second breathing and the beat times cannot? The original ECG and breathing signals, sampled 250 times a second (333 for one person), are on PhysioNet; the last cell of the starter notebook shows how to load them (for example, to find the heartbeats yourself, or to study the shape of each breath).

5. Interpretive statement (≤ 300 words, up to one figure)

Write a short statement, in your own words, that answers: How do the average heart rate and the heart-rate dynamics differ between young and elderly people, with numbers and uncertainties? How are the breathing dynamics and the heart-rate dynamics related, how sure are you, and what can a cross-correlation not tell you? What do the groups look like when all your quantities are considered together? How far do you trust these conclusions, from 40 people and your choices? What would you do next?

Base every claim on a result in your notebook, and quote the numbers. Put the statement at the end of your notebook, under the heading **Interpretive statement**, with a word count. It may include **one**

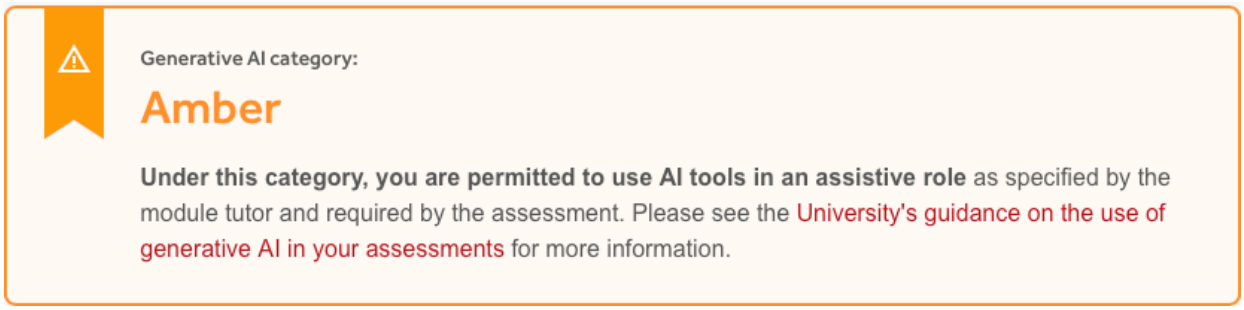
figure that carries your key result; an A+ needs one. If you prefer, submit the statement and its figure as a separate PDF instead, and say so at the end of the notebook.

6. Tips:

- Before submitting, make sure your notebook runs top to bottom with no edits: **Runtime → Restart session and run all**.
- Before each block of code, include a short Markdown cell saying what you are doing and why; after it, what the result means.
- Write functions for anything you do more than once (reading one person, the artefact rule, a spectrum, a band power), then loop over all 40 people.
- Visualisation counts for more in this exercise. Show every person where you can; give axes units; use colour consistently for the two groups; and make each figure answer one question. See `week2_code_and_plots.ipynb`.

7. Submission guidance, GenAI and help

- Submit on Moodle by 4pm, Friday 20 November 2026:
  - the **.ipynb notebook, already run**, with all outputs visible and your **statement** at the end (or as a separate PDF)
  - the **GenAI declaration** (mandatory). For an extension to the deadline, use the request form on the Part II Moodle page.
- GenAI:



  - may be used to assist with code development and debugging, and to explore analysis ideas. You must be able to explain every line you submit. Your statement, and the reasons for your choices in the notebook, must be your own words; GenAI may only correct spelling and grammar.
- Do ask questions in the labs.
- There will be a feedback session on Friday 27 November 2026, where we go through the solutions and discuss common issues. Marked work will be returned the following week.

8. Suggested plan (about 11 hours in total)

When	Suggested work	Time
Up to Lecture 6	Tasks 1–4, using Lectures 4–6 and the Week 4 and 5 labs (describing data, tests, spectra, resampling); Lecture 8 returns to Tasks 1, 2 and 4. Task 5 can start any time after Lecture 7.	6–7 h
Lectures 7 and 8, and the Week 6 lab	Finish Task 5, using Lecture 7 (PCA) and the Week 6 lab (k-means). Then decide on your extension from your Task 1–5 results; discuss it with a PGTA.	2–3 h + lab time
After the Week 6 lab, to the deadline (20 November)	Carry out and check the extension; write the statement; restart and run all; submit.	2–3 h

Marking grid

NB: At level 7, D and below is a fail. A+ is reserved for work where no improvement can be suggested.

University scale	Results (Tasks 1–5) 50%	Extension (Task 6) 15%	Interpretive statement 15%	Coding 8%	Visualisation 12%
Exceptional (1st Class) A+ [24]	Everything in Excellent, with every result confirmed by an independent check. Precision and uncertainties exactly as the evidence supports.	Goes beyond the obvious next step: finds and quantifies a subtler effect, with independent checks that agree. Could be handed to another scientist as it stands.	Concise and compelling, with one figure that carries the key result. Every claim quantified. Separates statistical from systematic uncertainty, states the limits of the conclusion and the most valuable next measurement.	Clean, reusable functions, tested on cases with known answers. Another student could reuse it without asking.	Figures that make the key result obvious at a glance, each designed for the point it supports.
Excellent (1st Class) A− A [18, 21]	All required results correct, with uncertainties, units and sensible precision. Methods justified and checked.	Clearly motivated by your own results. Appropriate, correctly done and checked independently. Outcome quantified and used in your conclusions.	A best estimate with its uncertainty. Explains, with your numbers, why results agree or differ. Judges how far to trust them. A specific next step.	Runs top to bottom. Functions for repeated work. Markdown explains why. Readable.	Clear, labelled with units, uncertainties shown, every person shown where possible, each figure used in the argument.
Good (Upper 2nd) B− B B+ [15, 16, 17]	Results correct, with minor slips. Limited justification or checking.	A sensible next step linked to your results, done correctly. Limited checking.	Mostly supported by results; describes more than it explains. Next step generic.	Runs. Mostly clear; some repetition or thin explanation.	All figures present and labelled; one weakness.
Satisfactory (Lower 2nd) C− C C+ [12, 13, 14]	Partly correct, or a method error. Uncertainties missing or misused.	Only loosely linked to your results, or partly incorrect. Outcome not quantified.	Mainly a description of what was done. Little explanation. Numbers without uncertainties.	Runs only after small fixes, or hard to follow.	Figures hard to read, or key elements missing.
Weak (3rd Class) D− D D+ [9, 10, 11]	Largely incorrect or incomplete. No uncertainties.	A token analysis with no stated reason, or wrong without awareness.	Conclusions not supported by, or contradicting, your results.	Does not run, or results cannot be traced to code.	Unlabelled or misleading figures.
Fail F4 F3 F2 F1 [0, 2, 4, 7]	Little or nothing usable.	Absent, or trivial.	Missing, irrelevant or largely incorrect.	No usable code.	No figures.